

Homework 9

Due: Wednesday, April 17, 2024

All homeworks are due at 11:59 PM on Gradescope.

Please do not include any identifying information about yourself in the handin, including your Banner ID.

Be sure to fully explain your reasoning and show all work for full credit.

Problem 1

For each of the statements below, determine if it is *always* true, *sometimes* true, or *never* true. Justify your answers. To justify an “always” or “never” answer, write a proof; to justify a “sometimes” answer, give one witness that makes the statement true and one that makes the statement false, explaining these judgments.

For example, the statement

Let $\mathbb{P}$ be a probability function on a sample space Ω . Then $\mathbb{P}[\omega] = \mathbb{P}[\omega']$ for every pair of outcomes $\omega, \omega' \in \Omega$.

is *sometimes* true. It is true for our classic “coin flip” probability space, where $\Omega = \{H, T\}$ with $\mathbb{P}[H] = \mathbb{P}[T] = \frac{1}{2}$. It is false for a “weighted die” probability space, where $\Omega = \{1, 2, 3, 4, 5, 6\}$ with $\mathbb{P}[1] = \mathbb{P}[2] = \mathbb{P}[3] = \mathbb{P}[4] = \mathbb{P}[5] = .1$ and $\mathbb{P}[6] = .5$.

- Let $\mathbb{P}$ be a probability function on a sample space Ω . The only event whose probability is 0 is the empty event $\emptyset \subseteq \Omega$.
- Let $\mathbb{P}$ be a probability function on a sample space Ω and let A and B be events with $\mathbb{P}[B] > 0$. Then $\mathbb{P}[A \mid B] \geq \mathbb{P}[A]$.
- Let $\mathbb{P}$ be a probability function on a sample space Ω and let A and B be events such that $\mathbb{P}[B] > 0$ and no outcome is in both A and B . Then $\mathbb{P}[A \mid B] = 0$.
- Let $\mathbb{P}$ be a probability function on a sample space Ω and let A and B be independent events with $\mathbb{P}[B] > 0$. Then $\mathbb{P}[B] = \mathbb{P}[A]$.
- Let $\mathbb{P}$ be a probability function on a finite sample space Ω and let R be a random variable on this probability space, taking values in $\mathbb{R}$. Then $\mathbb{E}[R]$ is infinite.

Problem 2

The Weather Channel's Thursday night weather report predicts a 60% chance that it will rain tomorrow (Friday).

In general, if it rains on a given day there is an 80% chance that it will rain the next day, but if it does not rain on a given day there is only a 10% chance that it will rain the next day.

- a. To answer the following questions, you will need to model the situation as a *probability space*. Read the questions below, and come up with the model you will use. What is the *sample space*? What are the *events* we will consider in parts b-d, expressed as sets of outcomes?
- b. What is the probability that it will rain on Saturday?
- c. What is the probability that it will rain on Sunday?
- d. What is the probability that it will rain at least once this coming weekend?
- e. Given that it will rain on Saturday, what is the probability that it rains tomorrow?

Problem 3

Let $D = \{1, 2, \dots, 9, 10\}$. Let $(\Omega, \mathbb{P})$ be a probability space, and let $R, S : \Omega \rightarrow D$ be random variables.

We say that R and S are *independent* random variables if for all $r, s \in D$, the events $R = r$ and $S = s$ are independent.

We say that R is *uniform* if for every $d \in D$, $\mathbb{P}[R = d] = \frac{1}{|D|}$.

Finally, we define the event $R = S$ to be the set of outcomes in Ω on which R and S take equal values:

$$R = S = \{\omega \in \Omega \mid R(\omega) = S(\omega)\}.$$

Suppose that R and S are independent and R is uniform. We claim that $\mathbb{P}[R = S] = \frac{1}{|D|}$. Intuitively, it seems whatever value S happens to take, R is just as likely to take that value as any other. But this is not a rigorous argument! Write a careful proof of this claim.

HINT: Can you write the event $R = S$ as a union of disjoint events?



Problem 4 (Mind Bender — *Extra Credit*)

This week's mindbender is a Lean question!

The problem can be found by navigating to `BrownCs22/Homework/Hw9.lean` in the directory browser on the left of your screen in Gitpod. The comment at the top of that file provides more detailed instructions.